

Hostel- Hunting using Fuzzy Logic

UI- NAVIGATION FLOW

1. Launch Application:

When the user runs the application using: `python app.py`

the homepage of the Hostel Hunter is displayed.

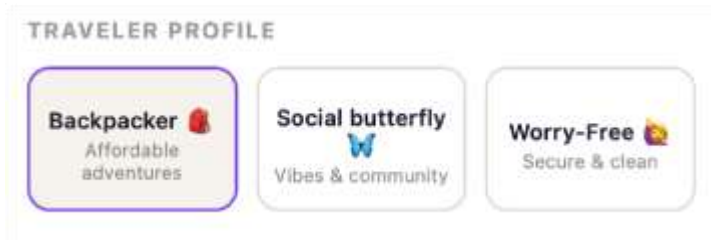
The screenshot displays the 'Hostel Hunter' application interface. On the left, the 'TRAVELER PROFILE' section includes three selectable traveler types: 'Backpacker' (All around adventures), 'Social butterfly' (Meet & community), and 'Worry-Free' (Secure & clean). Below these are four preference sliders: 'Nature' (30%), 'Experiences' (20%), 'Environment' (28%), and 'Safety' (22%). The 'Your Preferences' section features four sliders: 'Budget (max/night)' (50 €), 'Max Distance from Centre' (7.0 km), 'Min Safety Rating' (1.0 / 10), and 'Min Cleanliness' (1.0 / 10). Each slider has corresponding 'Low', 'Medium', and 'High' buttons with associated fuzzy logic values. The right side of the interface shows the 'Hostel Hunter Results' section with the text 'Find your perfect hostel with fuzzy logic' and a prompt to 'Set your preferences and click "Find Hostels" to get started'.

2. Enter User Preferences:

The user must select the traveller type:

Traveler Type:

- **Backpacker** - For travellers whose primary concern is staying within a budget while still enjoying local experiences and social interactions.
- **Social Butterfly** - For travellers who enjoy meeting new people and engaging in social activities.
- **Worry-Free** - For travellers who prioritize safety, cleanliness, and reliable accommodations above all other factors.



Preference Settings –

Set the preferences as per the requirement for below filters.

Budget, Maximum Distance from City Centre, Safety Requirement, Cleanliness Requirement, Social Importance, Wi-Fi Importance, Breakfast inclusion Preference, Nearest Station available Preference.

★ Your Preferences

Budget (max/night) 36 €

Low ($\mu=0.27$) Medium ($\mu=0.55$) High ($\mu=0.00$)

Max Distance from Centre 5.0 km

Near ($\mu=0.00$) Moderate ($\mu=0.25$) Far ($\mu=0.50$)

Min Safety Rating 5.0 /10

Low ($\mu=1.00$) Medium ($\mu=0.00$) High ($\mu=0.00$)

Min Cleanliness 6.0 /10

Low ($\mu=0.50$) Medium ($\mu=0.50$) High ($\mu=0.00$)

Social Vibe Importance 8 /10

Low ($\mu=0.00$) Moderate ($\mu=0.50$) High ($\mu=0.33$)

WiFi Importance 6 /10

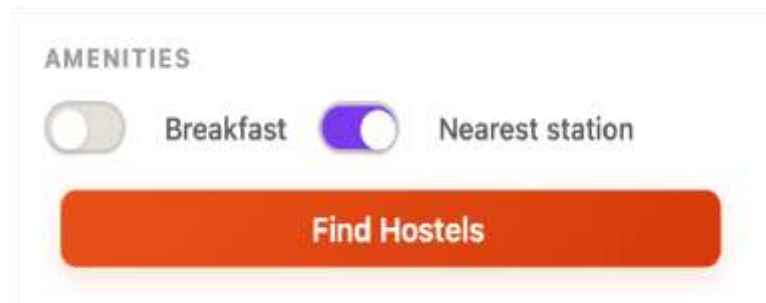
Low ($\mu=0.50$) Moderate ($\mu=0.50$) High ($\mu=0.00$)

AMENITIES

☒ Breakfast ☒ Nearest station

After setting all

preferences, click **“Find Hostels”**



3.Fuzzy Recommendation Processing:

The application sends the user inputs to the fuzzy logic engine.

Value Module

- Budget
- Price

Convenience Module

- Distance
- Wi-Fi
- Station Access

Experience Module

- Cleanliness
- Social Atmosphere

Safety Module

- Safety Rating

The system calculates a recommendation score for every hostel.

4.Hostel Ranking Generation:

All hostels are sorted according to their final fuzzy recommendation score.

The system generates:

- Ranked hostel list
- Recommendation scores
- Hostel details

5.Display Results:

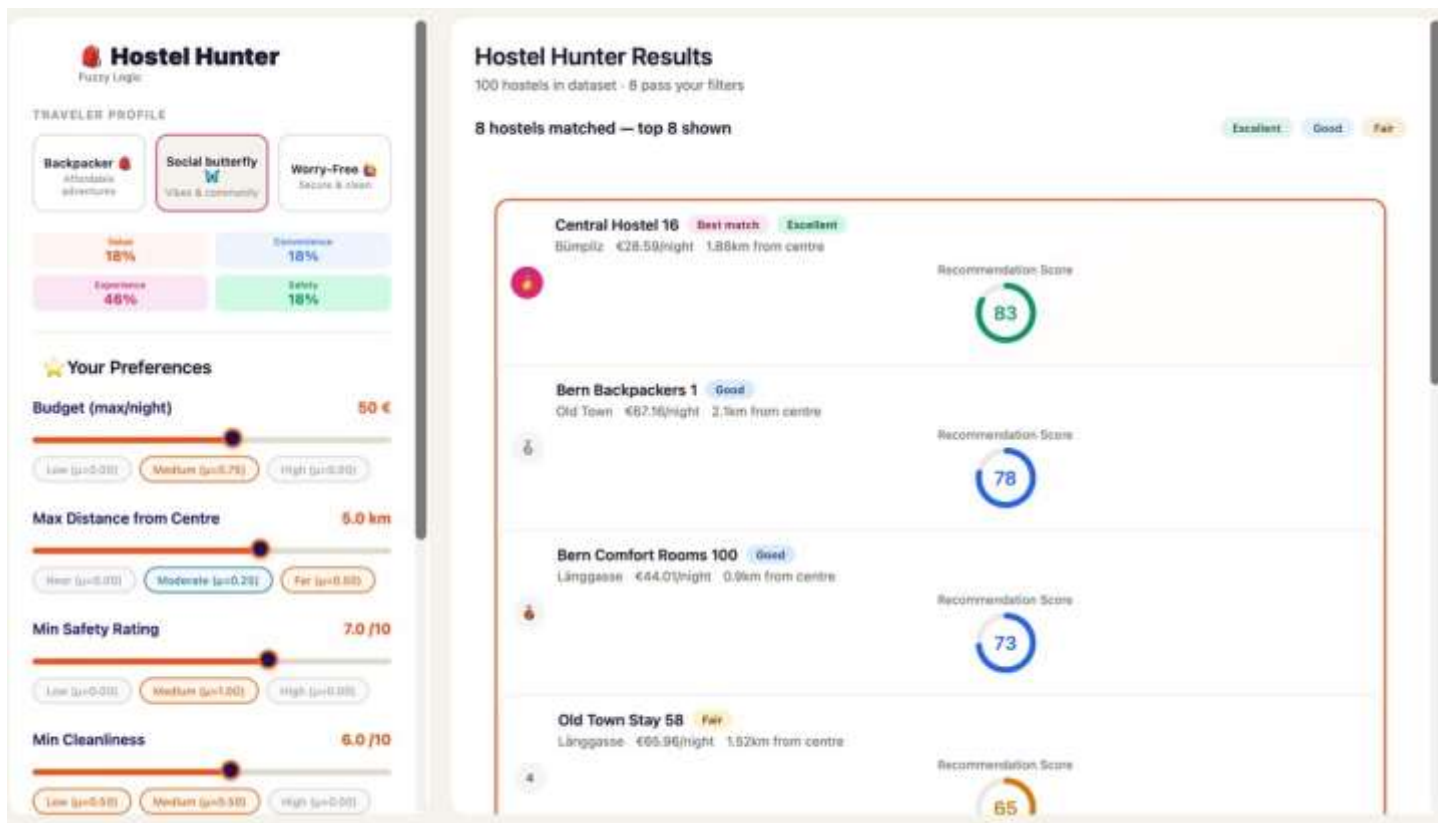
The results page displays:

Top Recommended Hostel -

The highest-ranked hostel based on user preferences.

Ranked Recommendations -

List of matching hostels sorted by score.



Hostel Information -

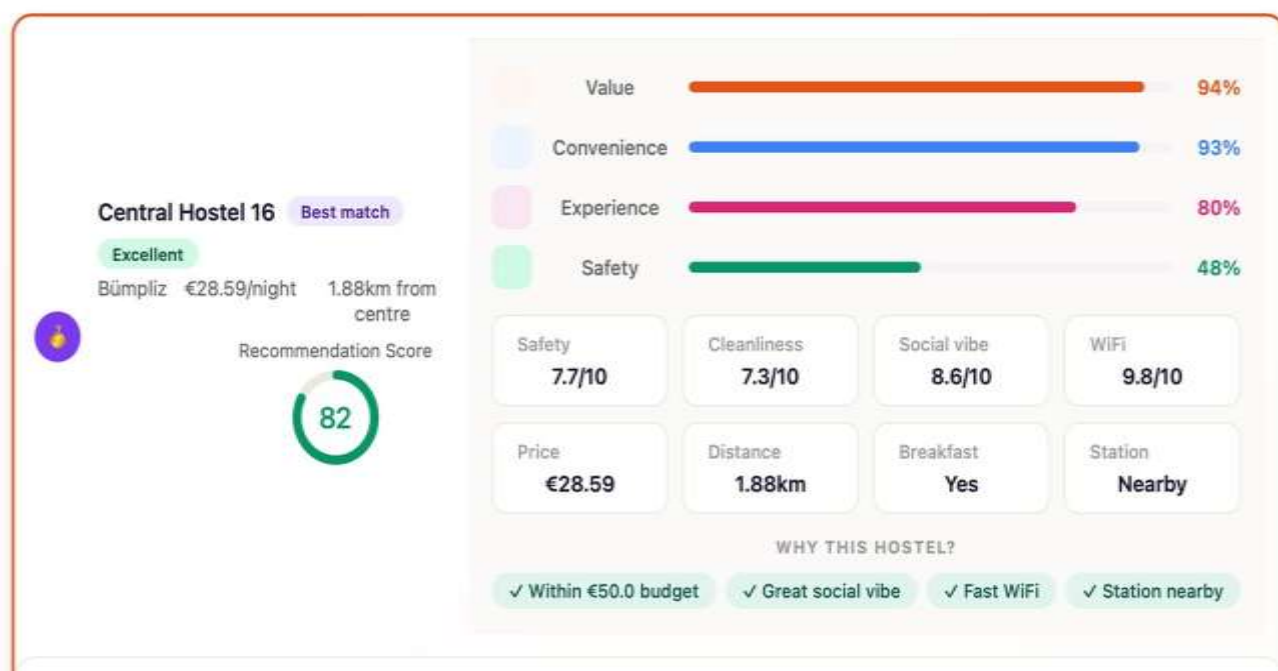
Price per night, Distance from city centre, Safety rating, Cleanliness rating, Wi-Fi quality, social atmosphere, Breakfast availability, Station accessibility

Hostel Hunter Results

100 hostels in dataset - 21 pass your filters

21 hostels matched — top 8 shown

Excellent | Good | Fair



6. Review Recommendations:

The user can:

- Compare hostels
- Review hostel scores
- Select the most suitable accommodation



